MIL-C-24643/9B
3 October 1986
SUPERSEDING
MIL-C-0024643/9A(SH)
8 November 1985
MIL-C-24643/9
28 September 1984

MILITARY SPECIFICATION SHEET

# CORD, ELECTRICAL, TYPE LSMRI

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the cord described herein shall consist of this specification and the latest issue of MIL-C-24643.

REQUIREMENTS:

Qualification required.

## Construction (non-watertight)

First - Copper conductor, class K stranding, uncoated (see table I for size).
Second - Ethylene propylene rubber or cross-linked polyethylene insulation (see table I for dimensions). Standard identification code, applied by method 3.
Third - The specified number of conductors cabled with a lay not greater than that specified (see table I).

TABLE I. Details.

| Military part no. M24643/9 | Type and size | Number of conductors | Conductor size AWG | Diameter over insulation | | Lay of conductors maximum (inch) | Conductor resistance per 1000 feet maximum (ohms) |
|---|---|---|---|---|---|---|---|
| | | | | minimum (inch) | maximum (inch) | | |
| -01UN | LSMRI-D-1 | 2 | 20(class K) | 0.068 | 0.075 | 2 | 11.05 |
| -02UN | LSMRI-D-2-1/2 | 2 | 16(class K) | .092 | .102 | 3 | 4.36 |
| -03UN | LSMRI-T-2-1/2 | 3 | 16(class K) | .092 | .102 | 3 | 4.36 |

(B) denotes changes.

AMSC N/A FSC 6145
DISTRIBUTION STATEMENT A Approved for public release; distribution unlimited



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EXAMINATION AND TESTS:

| | Requirements |
|---|---|
| **Basic electrical:** | |
| Conductor resistance - ohms/1000 feet at 25°C, maximum.. | (see table I) |
| Voltage withstand - volts, root mean square, minimum | |
| AWG 20 Class K..... | 1000 |
| AWG 16 Class K..... | 1500 |
| Insulation resistance - megohms/1000 feet, minimum..... | 100 |
| Conductor continuity..... | No failure |
| **Group A:** | |
| Visual and dimensional..... | No failure |
| **Group B:** | |
| Physicals (unaged) | |
| Insulation | |
| Ethylene propylene rubber | |
| Tensile strength - $lb/in^2$, minimum..... | 700 |
| Elongation - percent, minimum..... | 250 |
| Cross-linked polyethylene | |
| Tensile strength - $lb/in^2$, minimum..... | 1800 |
| Elongation - percent, minimum..... | 250 |
| **Group C:** | |
| Physicals (aged) | |
| Insulation | |
| Ethylene propylene rubber | |
| Air oven | |
| Tensile strength - percent of unaged, minimum..... | 75 |
| Elongation - percent of unaged, minimum..... | 75 |
| Cross-linked polyethylene | |
| Air oven | |
| Tensile strength - percent of unaged, minimum..... | 80 |
| Elongation - percent of unaged, minimum..... | 80 |

*QUALIFICATION INSPECTION:*

Qualification inspection shall include basic electrical, all of groups A, B, and C, plus the following:

| | |
|---|---|
| Acid gas equivalent - percent, maximum..... | 18 |
| Flame propagation..... | No failure |
| Toxicity index, maximum..... | 1.5 |
| (B) Smoke index, maximum..... | 45 |
| Heat distortion - percent of unaged, maximum..... | 30 |



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UNIT ORDERING LENGTHS:

| Type | Size | Feet (nominal) |
|---|---|---|
| LSMRI-D- | 1 | 600 |
| LSMRI-D- | 2-1/2 | 600 |
| LSMRI-T- | 2-1/2 | 500 |

NOTE: Not for Air Force use.

Custodians:
Army - MI
Navy - SH

Preparing activity:
Navy - SH
(Project 6145-1005-09)

Review activities:
Army - AV, CR, ER
Navy - EC
DLA - IS

User activities:
Army - ME, AR, AL
Navy - CG



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